Viraj Mankani

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EDUCATION

National Institute of Technology, Tiruchirappalli **CGPA: 8.10**
Bachelor of Technology in Computer Science and Engineering *Aug. 2023 – July 2027*

EXPERIENCE

Tata Communications – Certificate *May 2026 – July 2026*
Project Trainee, Software Engineering *Mumbai, India*

- Architected high-throughput backend microservices in Python and FastAPI with async dependency injection, Pydantic v2 schemas, and Async SQLAlchemy 2.0; engineered OAuth2/OIDC Microsoft SSO and local JWT validation across 30+ async REST endpoints for 100+ daily enterprise users.
- Engineered an asynchronous task and event processing engine using Redis and *asyncio*, implementing distributed locks, idempotency keys, and multi-tier cache-aside layers with selective TTL invalidation, reducing database query load by 60% and reducing p95 latency from 180ms to 25ms.
- Built custom ASGI telemetry middleware for distributed *X-Correlation-ID* request tracing, token bucket rate-limiting, and structured JSON metrics in Prometheus and Grafana; authored zero-downtime Alembic migrations and maintained 92% test coverage on GitLab CI/CD.

Lyceum Development Centre – Certificate *May 2025 – July 2025*
Machine Learning Intern *Chennai, India*

- Built high-throughput data ingestion pipelines in Python, streaming and parsing 300+ enterprise RFP documents per month into structured JSON schemas for downstream RAG pipelines, eliminating manual processing overhead by 80%.
- Stood up ChromaDB vector database infrastructure holding 5M+ embeddings to power LangChain RAG pipelines, improving query retrieval precision by 20% and reducing document processing latency by 6x.
- Deployed a production semantic search service across 1,800+ QA pairs using LangChain and transformer embeddings, sustaining sub-3s inference latency, and shipped an interactive Gradio user dashboard.

PROJECTS

NginZ – Distributed API Gateway & Service Coordinator – GitHub *March 2026*
Node.js, Express.js, TypeScript, Nginx, Docker, Redis, PostgreSQL, Prometheus, Grafana

- Architected a distributed cloud-native API gateway on Nginx and Node.js/Express, coordinating 10 Dockerized instances with automatic Redis-backed service discovery, SIGTERM graceful deregistration, and zero downtime during simulated central auth outages via local JWT validation.
- Prevented cascading service degradation by engineering a Redis-backed Token Bucket rate limiter and per-service Circuit Breakers with exponential backoff and jitter; validated 1000 RPS sustained throughput with Prometheus metrics and 5 live Grafana dashboards.

Product Search System – GitHub *March 2025*
MEAN Stack (MongoDB, Express.js, Angular, Node.js), Elasticsearch, TypeScript, Docker

- Built a full-stack e-commerce search platform on the MEAN stack, integrating Elasticsearch-powered full-text search, autocomplete, and relevance ranking with faceted multi-attribute filtering on a responsive Angular frontend.
- Structured a scalable catalog query layer with MongoDB schemas and Express REST APIs for sub-second responses; containerized the entire application with Docker for consistent reproducible deployment.

Shopizer E-Commerce Cloud Deployment – GitHub *April 2026*
Java, Spring Boot, AWS (EC2, ALB, RDS, ECR, IAM, S3), Docker, MySQL, Maven

- Containerized a Java Spring Boot e-commerce monolith into Docker services and deployed on AWS with an Application Load Balancer and RDS MySQL database, implementing automated multi-stage CI/CD deployment pipelines with zero downtime.
- Engineered secure cloud infrastructure with custom AWS IAM policies, isolated VPC subnets, and RDS automated backup policies; hardened Linux instances with security groups and POSIX access controls, achieving 99.9% uptime across stress testing.

TECHNICAL SKILLS & ACHIEVEMENTS

Languages: Python, C, C++, Java, TypeScript, JavaScript, SQL, Bash, HTML/CSS
Backend & Frameworks: FastAPI (Async, Depends), Asyncio, Node.js, Express.js, Spring Boot, Async SQLAlchemy 2.0, Pydantic v2, REST APIs, Microservices, WebSockets, JWT, OAuth2/OIDC, RBAC
ML Frameworks & Applied AI: PyTorch, TorchVision, Vision Transformers (ViT), LangChain, ChromaDB (Vector DB), Scikit-learn, SISA Machine Unlearning, NumPy, Pandas, Gradio, Streamlit
Databases, Cloud & DevOps: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, AWS (EC2, ALB, RDS, S3, IAM), Docker, Kubernetes, Nginx, Prometheus, Grafana, CI/CD, Linux, Git
Competitive Honors: JEE **AIR 950** (16 Lakhs) • MHT-CET **State Rank 1** • 500+ DSA (LeetCode, GFG) • 1303 on Codeforces